RECAP. THEMES: &CASE EXAMPLE.

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COURSE OVERVIEW

THREAD MODELING

CYBER ATTACKS AND ADVISORY METHODS

WEB SECURITY. OWASP 10.

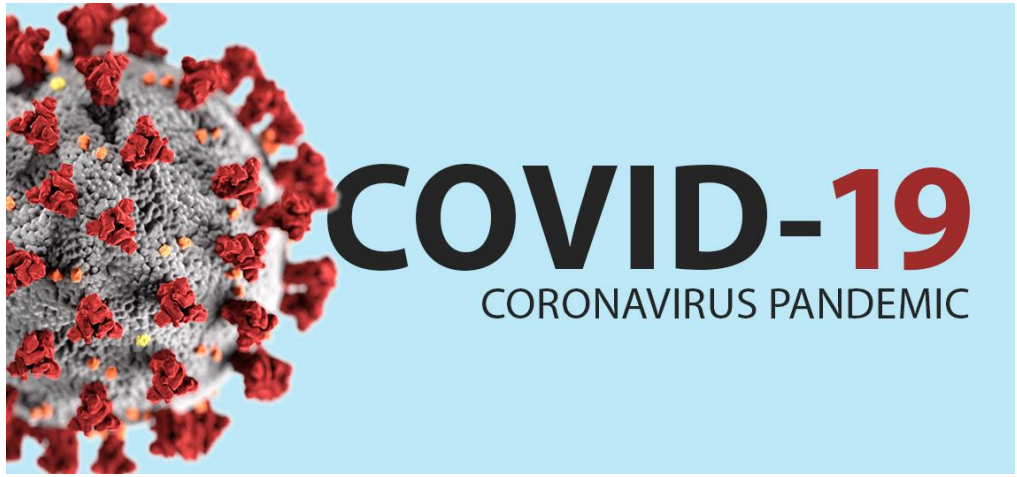
ENCRYPTION AND SECURE COMMUNICATION

IDENTITY AND AUTHENTICATION

NEW DEVELOPEMENTS. OSINT.

THREAT MODELING

- Threat modeling is a process used to identify security and privacy risks
- If you remember there were Four Key Questions
- It helps recognize risks early
- Anyone concerned about privacy, safety, or security



https://www.threatmodelingmanifesto.org/

CYBER KILL CHAIN, RULES AND DEFENSE

- It describes the 7 steps attackers follow when performing a cyber attack.
- This helps organizations to stop these attacks before any damage is done.
- GRC is a framework combines all three to ensure a company is actually following the law and identifying the right risks.
- Examples of simple safety practices include:

1. Use unique passwords
2. Enable 2FA.

https://www.microsoft.com/en-us/security/business/security-101/what-is-cyber-kill-chain#:~:text=The%20cyber%20kill%20chain%20defines,and%20how%20they%20evade%20detection.
https://www.legitsecurity.com/aspm-knowledge-base/grc-in-cybersecurity

WEB SECURITY AND OWASP 10

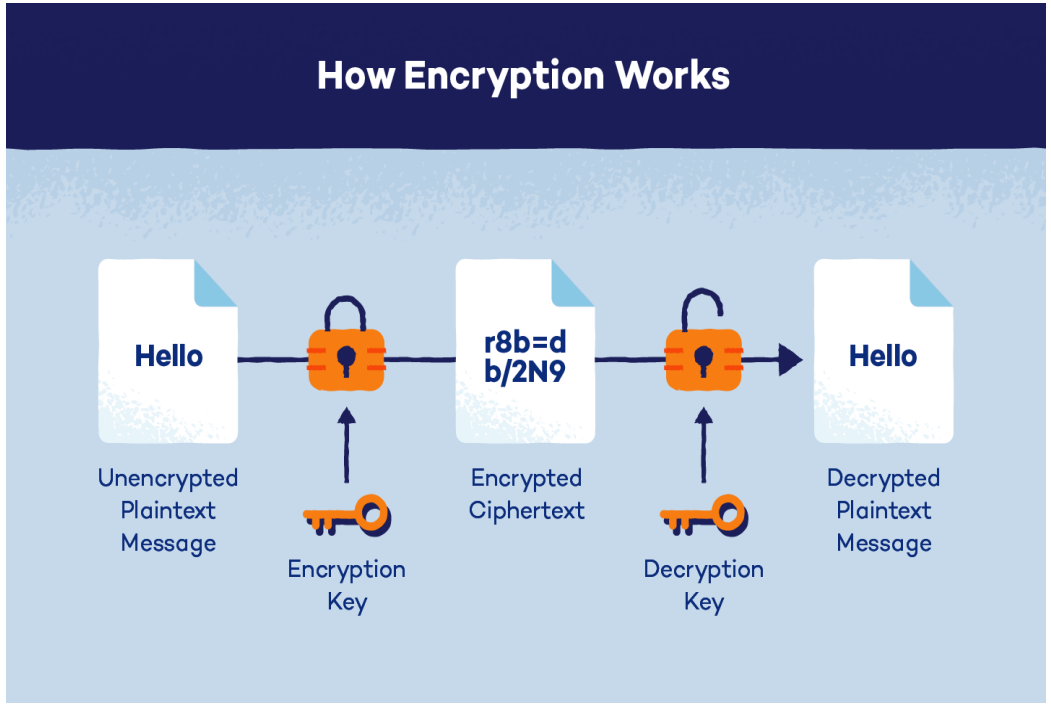
- **OWASP** is a foundation that works to improve software security.
- It provides us with a list of 10 most critical web application security risks.

1. Broken Access Control
2. Security Misconfiguration
3. Vulnerable and outdated components

- **Web security** focuses on protecting websites and web applications from cyber attacks.
- It aims to protect user data, systems, and services

OWASP TOP 10

https://owasp.org/Top10/2021/A00_2021_Introduction/
https://www.proofpoint.com/us/threat-reference/web-security



ENCRYPTION AND COMMUNCATION SECURITY

- Encryption is the process of converting plain text into cyphertext
- Sensitive information is protected from unauthorized access.
- Symmetric Encryption uses one shared key to encrypt and decrypt data
- Asymmetric Encryption uses two keys a public key and a private key.

https://www.ibm.com/think/topics/encryption#17494633

FOR SECURE COMMUNICATION

- **GPG**

1. A tool that helps you **encrypt** and **sign** your data.

- A **public key** (which you can share with anyone).
- A **private key** (which is kept secret and only for you to use).

- **SSH**

1. SSH is a protocol used to **securely connect to remote computers**.

- Secure remote login
- Secure file transfers (SFTP & SCP)

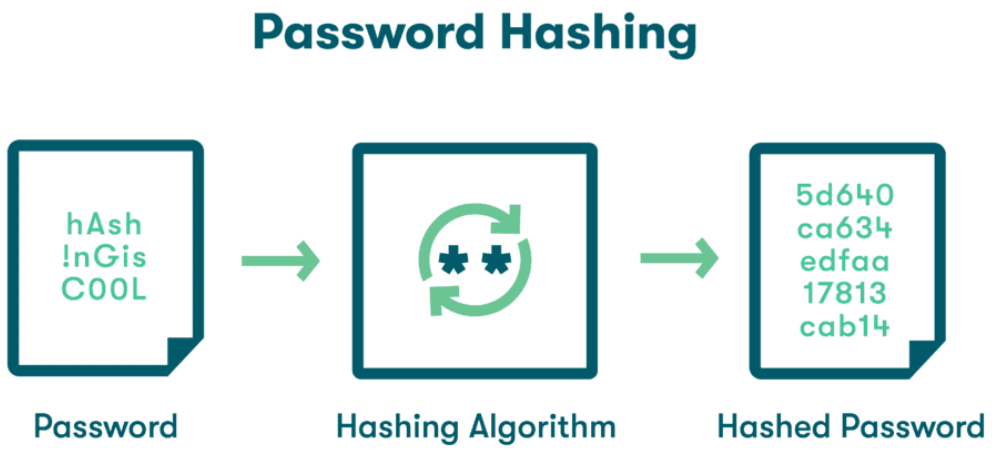
https://terrazone.io/what-is-gpg/

https://www.encryptionconsulting.com/everything-you-need-to-know-about-ssh/

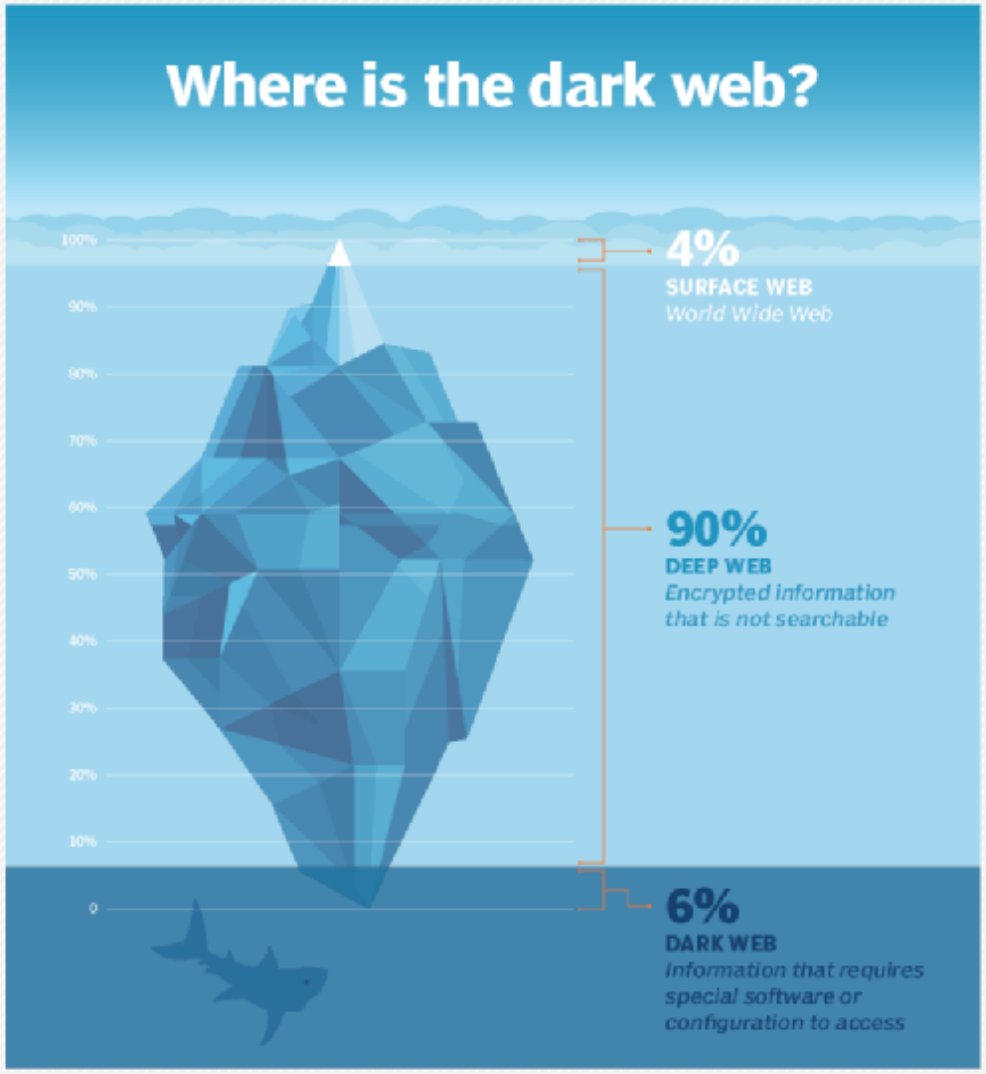
HASHES, IDENTITY, PASSWORDS AND AUTHENTICATION

- Hash is a function that converts data into a string of characters.
- It is a one - way function, meaning it can not be easily reversed back to original.
- It protects user's passwords even if a database is stolen.
- Authentication includes several checks for identity like:

1. Smartphones
2. Face IDs
3. MFA



https://www.infobip.com/glossary/password-hashing#how-is-password-hashing-used-to-grant-access



OSINT & Darknet

- **OSINT** means gathering publicly available data to analyze about any threats.
- It is often used in cybersecurity and by intelligence agencies.
- **Darknet** is a part within the internet that is not easily accessible. It require special software to access.
- Many activities on the Darknet are illegal.
- **TOR** is a network that can used to gain access to the Darknet.
- It protects user's IP address through multiple encrypted layers.

https://www.ibm.com/think/topics/osint
https://en.wikipedia.org/wiki/Darknet
https://en.wikipedia.org/wiki/Tor_(network)

REAL LIFE CASE EXAMPLE

- Happened in 2013
- Retail company Target suffered a major cyberattack.
- Hackers stole credit card information for about 40 million customers
- Even personal data of 70 million people was compromised/exposed.
- Cybercriminals were growing
- Major security shifts.
- They suffered a heavy financial and reputational blow out of nowhere

How did it take place?

- Attackers first targeted a **third-party contractor** which was "Fazio Mechanical Services".
- Hackers obtained credentials used by the vendor to access a Target web application.
- Once they gained access malware was installed on Target's point-of-sale systems (cash register systems).
- This allowed them to capture sensitive customer information, including payment card details and personal data.

Security problems

- Broken access control, attackers moved inside the network.
- Weak network segmentation, attackers accessed sensitive systems.

Lessons learned: Indeed, a wake-up call

- Network segmentation
- Monitor and respond to security alerts
- Secure third-party vendor access

https://frameworksecurity.com/post/the-target-breach-a-historic-cyberattack-with-lasting-consequences
https://redriver.com/security/target-data-breach

Thank you!